

Stock market anomalies: What can we learn from repurchases and insider trading?

John E. Core

Email: jcore@wharton.upenn.edu

Phone: (215) 898-4821

Wayne R. Guay

Email: guay@wharton.upenn.edu

Phone: (215) 898-7775

Scott A. Richardson

Email: scottric@wharton.upenn.edu

Phone: (215) 898-2063

Rodrigo S. Verdi

Email: rverdi@wharton.upenn.edu

Phone: (215) 898-7789

All authors are at the Wharton School, University of Pennsylvania
1300 Steinberg Hall-Dietrich Hall

First Version: April 20, 2004

Current Version: July 30, 2005

We appreciate the comments and suggestions of Peter Easton (editor), Joseph Gerakos, Theodore Goodman, Jeffrey Ng, Jonathan Rogers, Tjomme Rusticus, Sarah Zechman, two anonymous referees, and seminar participants at the University of Pennsylvania and 2004 European Accounting Association Meeting.

Stock market anomalies: What can we learn from repurchases and insider trading?

Abstract

We examine whether managers' trading decisions (both at a firm and personal level) are correlated with trading strategies suggested by the operating accruals and the post-earnings announcement drift (SUE) anomalies. We discuss advantages and disadvantages of the use of managerial trading activity to infer managers' private valuation about their own securities. Our results provide corroborative evidence for the accruals anomaly, i.e., managers' repurchase and insider trading behavior varies consistently with the information underlying the operating accruals trading strategy. On the other hand, we do not find corroborative evidence for the SUE anomaly.

An extensive literature has developed challenging the assumption of market efficiency (for a survey of this literature see Kothari, 2001 and Lee, 2001). Collectively these papers document empirical regularities consistent with investor under or over reaction to publicly available information. Although these anomalies appear to have withstood a barrage of robustness tests, the findings are not without criticism. Schwert (2003) notes that many anomalies occur during specific time periods or within particular selected samples that cannot be readily generalized or implemented on an ex ante basis. Similarly, Fama (1998) documents that long-term return anomalies are sensitive to empirical methods, in that the abnormal returns are sensitive to models for expected (normal) returns and statistical techniques. Ball (1992) and Kothari (2001) also discuss the robustness of return anomalies.

This paper adds to this literature by examining whether managers' trading decisions (both at a firm and personal level) are correlated with trading strategies suggested by the operating accruals and the post-earnings announcement drift (SUE) anomalies. We investigate these two anomalies because they seem especially suited for the purposes of this paper. Given the role that management plays in the financial reporting process, management are uniquely informed about financial reporting and may be in the best position to observe pricing deviations from fundamental value. The accruals anomaly, in particular, hinges on the investor inability to recognize the differential persistence of accruals and cash flows, and managers are at advantaged position to understand the firm's accruals process.

Our research design is based on empirical evidence that shows executives are well-informed about their firms' future expected cash flows and the cost of capital, and

use this information, at least at the margin, in their decisions about share repurchases and individual trading. For example, Ikenberry, Lakonishok and Vermaelen (1995) and D'Mello and Shroff (2000) find that repurchase decisions are influenced by managers' perceptions about stock price undervaluation; Seyhun (1992) finds that insider trading is consistent with managers taking advantage of deviations between stock prices and fundamental values; and Rozeff and Zaman (1998) and Beneish and Vargus (2002) link insider trading to misvaluations based on the market-to-book ratio and operating accruals.

Consistent with this evidence, we assume that executives, as insiders, are well-informed about whether their own equity is mispriced, i.e., whether current stock price diverges from fundamental value, and that they use this information at the margin in making share repurchase and insider trading decisions. Further, we assume that the information set available to insiders encompasses any mispricing that can be identified by anomalous trading strategies. Therefore, if the trading strategies based on operating accruals and post-earnings announcement drift do, in fact, identify equity mispricing, then we expect that insiders' share repurchase and individual trading decisions will be correlated with the positions recommended by these trading strategies.¹ We note that our tests do not require the executives to be aware of, or follow, any specific anomalous trading strategy. For our purposes, it is sufficient that the executives are skilled at identifying mispricings in their equity (regardless of the source of the mispricing), and that mispricing influences marginal trading behavior.

The use of managerial trading activity to infer managers' private valuation about

¹ For example, suppose that a firm with very low accruals is identified by the accruals trading strategy as being undervalued and that executives of this firm use their inside knowledge of expected future cash flows to estimate the true equity value. If the executives believe the stock is undervalued, at the margin, they will buy shares on personal and firm accounts more heavily, thereby corroborating the accruals trading strategy.

their own securities is a unique feature of our research design. We are interested in the extent to which the information managers use to generate abnormal returns is correlated with the “anomalous” public information used to form trading strategies according to the accruals and SUE anomalies. If we find evidence consistent with managers’ repurchase and insider trades coinciding with trades suggested by these anomalies, this supports the mispricing explanations offered for the anomalies. If, on the other hand, we find that the repurchase or insider trading activity is uncorrelated with the trading suggested by the anomalies, then the inferences we draw are constrained by the joint hypothesis nature of our research design. That is, failure to reject the null hypothesis could be due to (1) managers lacking the necessary sophistication to understand when their stock is mispriced, (2) the anomalies examined do not in fact measure mis-pricing, or (3) managers perceive prohibitive trading costs or risk associated with trading on the relevant mispricing. For example, if managers buy stocks based on the anomalies, they would have to take on additional firm-specific risk beyond the large firm-specific risk they already bear through their human capital and equity investment in the firm. When they do this, they are outside the typical diversified hedge-portfolio approach proposed by trading strategies. Distinguishing between these alternatives is difficult, and ultimately the interpretation of a non result depends on the readers’ priors with respect to managers’ knowledge of mispricing, trading costs, and the existence of the anomalies.

Using measures of operating accruals (Sloan, 1996) and unexpected earnings (Bernard and Thomas, 1989 and 1990), we group firms into ten equal-sized portfolios consistent with the trading strategies suggested by the anomalies literature. We then examine the level of abnormal share repurchases and insider trading for these portfolios

in the period leading up to, and following, the quarter in which we measure operating accruals and unexpected earnings. We use abnormal levels of stock repurchases and insider trading in order to mitigate omitted correlated variables concerns arising from the influence of expected factors affecting repurchase and/or insider trading decisions.

Our results provide corroborative evidence for the accruals anomaly, i.e., managers' repurchase and insider trading behavior varies consistently with the information underlying the operating accruals trading strategy. We document that low (high) accruals firms repurchase more (less) shares, and managers of low (high) accruals firms buy more (less) shares on their personal accounts. On the other hand, we do not find corroborative evidence for the SUE anomaly.

The fact that we find corroborating evidence for the accruals anomaly but not the SUE anomaly suggests that under the assumption that the SUE anomaly truly captures mispricing, trading on information of the type embedded in SUE is more costly to managers than trading on information of the type embedded in accruals. Furthermore, given that our research design has sufficient power to corroborate an anomaly that relies on financial statement information (i.e., accruals), future research could exploit this design to examine the validity of other financial-statement-based "anomalies".

Our findings on firms' repurchase choices are related to a literature that examines how market mispricing affects managers' real investment and financing decisions. For example, Polk and Sapienza (2004) find a positive relation between real investment and equity overvaluation as measured by accounting accruals. Ang and Cheng (2003), Dong, Hirshleifer, Richardson and Teoh (2005) and Rhodes-Kropf, Robinson and Viswanathan (2003) find that firm-level investment activity (and in particular M&A activity) varies

with the extent of firm level misvaluation. In particular, Dong et al. (2005) focus on pre-takeover ratios of residual income model value to price, and find that misvaluation of both bidders and targets influences the means of payment chosen, the mode of acquisition, the premia paid, target hostility to the offer, the likelihood of offer success, and bidder and target announcement period stock returns. Finally, Baker and Wurgler (2002) provide evidence that firms' capital structure is consistent with cumulative attempts to time the equity market and take advantage of market misvaluations.

Although these papers and our paper test similar joint hypotheses about misvaluation and managers' understanding of this misvaluation, our paper differs with respect to the research design. For example, because transactions costs associated with share repurchases and managerial trading are likely to be smaller than those for investment, we expect that managers can more easily adjust their repurchase and inside trading activity to take advantage of transient mispricing. On the other hand, constraints on insider trading activities, like insider trading restrictions on trades prior to major news events, and reputation costs may reduce the power of tests such that a failure to reject the null hypothesis may render the results inconclusive. In order to mitigate this last concern, we measure insider trading activity before and after the accounting information is made public, such that any trading activity based on public information would make violations of insider trading laws less likely.

The remainder of the paper proceeds as follows. Section 1 summarizes the anomalies we investigate. Section 2 develops expectation models for share repurchase and insider trading activity. Section 3 describes the sample selection and research design. Section 4 presents our main results and Section 6 provides a summary and concluding

remarks.

1. Operating Accruals and Post Earnings Announcement Drift

In this section we briefly summarize the prior literature that established the operating accruals and the post earnings announcement drift anomalies we investigate.

1.1. Accruals Anomaly

Sloan (1996) finds that stock prices do not properly reflect the information about future earnings contained in current earnings. His results suggest that investors fail to understand that the accruals component of earnings is less persistent than the cash flow component of earnings. In particular, a portfolio strategy that takes a long (short) position in firms reporting low (high) levels of operating accruals relative to cash flow, generates size-adjusted abnormal returns of 10.4% on an annual basis (for a large sample of firms over the 1963-1991 period).

The accruals anomaly is robust to different specifications such as the use of investing and financing accruals (Richardson et al., 2005), different countries (Pincus et al., 2003), and the use of quarterly data (Collins and Hribar, 2000), among others. Overall, the results are robust and consistent with Sloan's (1996) findings and suggest that future stock returns could be predicted due to investors *overreaction* to the persistence of the accruals component of earnings.

1.2. Post Earnings Announcement Drift (SUE) Anomaly

The SUE anomaly, first reported by Ball and Brown (1968) and subsequently documented in many other studies, suggests that stock prices continue to rise in the year following a positive earnings surprise and continue to fall in the year following a negative

earnings surprise.² Bernard and Thomas (1989, 1990) investigate competing explanations for the SUE effect and find results consistent with a delayed response of stock prices to new information in earnings announcements. They find that an arbitrage portfolio that takes a long position in firms reporting high standardized unexpected earnings (SUE) and a short position in firms reporting low SUE generates abnormal returns of approximately 8-9% in the first quarter after the earnings announcement (Table 5, Bernard and Thomas, 1990). Likewise, Collins and Hribar (2000) find an average two-quarter abnormal return of about 6.88% for the SUE hedge portfolio strategy. The results suggest that the market *underreacts* to the earnings announcement and responds only gradually to new information.

2. Measures of Share Repurchases and Insider Trading

In this section we define our measures of both repurchase and insider trading activity, and develop models for the expected levels of these activities. As we note below, a substantial body of evidence documents that share repurchases and insider trading varies with firm characteristics. To increase the power of our tests, we control for cross-sectional variation in managers' trades due to these economic determinants, and we use the resulting estimates of unexpected or residual insider trading and share repurchases in our tests. As a standard caveat, our expectation models may reflect mis-specification in both the functional form (linear) and the set of included/excluded independent variables. We also note a potential concern that some of the determinants of repurchases and insider trading may be correlated with the proxies for mispricing identified by the anomalies. Controlling for these determinants may affect our ability to find a relation between the

² See Jones and Litzenberger (1970), Foster et al. (1984), Chan, Jegadeesh and Lakonishok (1996), and Ball and Bartov (1996), Bushee and Raedy (2003), among others.

anomalies and managers' trades. Therefore, for robustness, we run all of our tests using unadjusted share repurchases and insider trading variables (results discussed in section 5).

2.1. Share Repurchases

Managers generally cite undervaluation as a reason for repurchasing shares (Brav et al., 2005), and empirical evidence documents that managers repurchase more heavily when they believe the stock is undervalued and repurchase fewer shares when they perceive the stock to be overvalued (Ikenberry, Lakonishok and Vermaelen, 1995, 2000). Therefore, given our assumption that managers' information set subsumes the information set academics use to structure anomalous trading strategies, we expect that if a particular anomaly represents the deviation of stock price from fundamental value, firms will adjust their repurchase programs to benefit from the temporary mispricing inherent in the anomaly.

Actual values of share repurchases are difficult to measure precisely for U.S. firms (Stephens and Weisbach, 1998). We measure *Repurchase* as the percentage change in split-adjusted shares outstanding over a six-month period as reported by CRSP, multiplied by -1 so that the measure is increasing in the number of shares repurchased. For example, if a firm XYZ has 50 shares outstanding at the end of month $m-4$, where month m is the ending month of a given fiscal quarter t , and 45 shares outstanding at the end of month $m+2$, *Repurchase* equals +10% (i.e., $-1 * (45-50)/50 * 100$). Negative values of this measure indicate that new shares were issued during the six-month period around quarter t .

One concern with the *Repurchase* measure is that it is inclusive of all transactions that decrease or increase the number of outstanding shares. As such, it includes follow-on

offerings that require share registrations and are costly and time-consuming, and thereby unlikely to be used to take advantage of temporary equity mispricing. To increase the power of our tests, we would like to limit our analysis to share repurchases and share re-issuances out of treasury that the firm can execute inexpensively and on short notice. Unfortunately, it is not possible to accurately identify the component of the change in shares outstanding that relates to follow-on offerings. Instead, we attempt to identify and reduce the influence of these offerings by examining the distribution of the six-month change in treasury stock. We do this by computing the two-quarter change in the dollar amount of shares held in treasury (data #98) and deflating it by the market value of total shares outstanding at the start of the period for all firm-quarter observations in our sample.³ The first percentile of this distribution is -3.85% suggesting that the large re-issues out of treasury are about 4 percent of outstanding shares. We then winsorize 10.83% of our sample where *Repurchase* is more negative than -3.85%, i.e., if the actual value is less than -3.85% of shares outstanding, then we replace it with -3.85%. This allows us to remove follow-on equity offerings from our net repurchase measure yet still retain re-issues from treasury.

We estimate abnormal share net repurchases as the residual from an OLS prediction model that includes proxies for the economic determinants of share repurchases largely based on Dittmar (2000). We expect firms with high cash balances to repurchase more shares, and firms with high capital expenditures and high debt to repurchase fewer shares. *Cash* is the ratio of cash and cash equivalents (data #36) to the

³ All data items refer to the Compustat quarterly files.

book value of total assets measured at the end of quarter t-2.⁴ *Cap. Expenses* is the ratio of capital expenditures (data #90) to the book value of total assets, and *Debt* is the ratio of long-term debt (data #51) plus the long-term debt included in current liabilities (data #45) to the book value of total assets, all measured in quarter t-2. We also include *Dividend Yield* as the dividend-per-share (data #16) divided by the mean closing price in the quarter ((data #12 + data #13 + data #14)/3). It is calculated for the six month period ending in the end of quarter t-2. We do not predict the sign for this variable because it is uncertain whether dividends and share repurchases are complements or substitutes (Grullon and Michaely 2002).

We control for *Size* using the natural logarithm of the book value of total assets (data #44) at the end of quarter t-2. We expect that small firms have greater information asymmetries and should have more opportunities to repurchase shares for misvaluation purposes. However, Dittmar (2000) finds that large firms are more likely to repurchase shares, so we do not make any prediction about the sign of the estimated coefficient for this variable. We also include fiscal quarter and year dummies to control for seasonality in firm share repurchase volume. We include industry indicator variables based on firms' two-digit SIC code.⁵ Finally, we include a measure of lagged *Repurchase* because we expect that repurchase patterns are persistent.⁶

$$\begin{aligned} Repurchase_t = & \beta_0 + \beta_1 * Repurchase_{t-2} + \beta_2 * Size_{t-2} + \beta_3 * Cash_{t-2} + \beta_4 * Debt_{t-2} + \\ & \beta_5 * Cap. Expenses_{t-2} + \beta_6 * Dividend Yield_{t-2} + \sum \beta_i * Year_i + \end{aligned}$$

⁴ As we discuss in more detail below, we measure the six-month period to end two months after a quarter t. Accordingly, financial variables relevant to the repurchase decision are measured at quarter t-2, which ends two-months prior to the computation of repurchase (and insider trading) data.

⁵ We delete firms in two-digit SIC codes that have an average of less than two observations per quarter.

$$\sum \beta_j * Quarter_j + \sum \beta_m * Industry_m + \varepsilon. \quad (1)$$

2.2. *Insider Trading*

We obtain insider trading data from Thomson Financial Securities Data. We focus on open market transactions and, following Beneish and Vargus (2002), our measure of insider trading is limited to transactions of the top five executives (CEO, CFO, COO, President, and Chairman of board corresponding to relationship codes of “CB”, “CEO”, “CFO”, “CO” and “P” on the Thomson Financial database) because previous papers document that these top executives are more likely to possess private information (Seyhun, 1986).

Following Rozeff and Zaman (1998) and Piotroski and Roulstone (2005), we measure *Insider Trading* using the firm’s purchase ratio, defined as

$$Insider\ Trading_{i,t} = Buy_{i,t} / (Buy_{i,t} + Sell_{i,t}) \quad (2)$$

where $Buy_{i,t}$ ($Sell_{i,t}$) is the number of shares purchased (sold) by the top five executives of firm i during the six-month period ending two months after quarter t (i.e., the same 6 month period examined for stock repurchases).

Similar to the share repurchases model, we estimate abnormal insider trading as the residuals from an OLS prediction model that includes variables we expect are correlated with insider trading but not correlated with the variables underlying the anomalies. Prior findings suggest that small firms have relatively more purchases than sales (Seyhun, 1986 and Rozeff and Zaman, 1988), and we control for *Size* with the

⁶ Our inferences are unaffected if we do not control for lagged repurchases.

natural logarithm of the book value of total assets (data #44) at the end of quarter t-2. We include fiscal quarter and year dummies to address seasonality in insider trading behavior, and two-digit SIC industry dummies to control for industry effects. As with the repurchase model we also include a measure of lagged *Insider Trading* to capture the normal propensity of insiders at the firm to sell/buy stock.

$$\begin{aligned} Insider\ Trading_t = & \beta_0 + \beta_1 * Insider\ Trading_{t-2} + \beta_2 * Size_{t-2} + \sum \beta_i * Year_i + \\ & \sum \beta_j * Quarter_j + \sum \beta_m * Industry_m + \varepsilon. \end{aligned} \quad (3)$$

Our main hypothesis is to test whether shares repurchase and insider trading activity are associated with trading strategies suggested by the operating accruals and SUE anomalies. Thus we use the residuals from Equations 1 and 3 as our measures of *Ab. Repurchase* and *Ab. Insider Trading*, and we correlate these variables with our measures of operating accruals and SUE.

3. Sample Description, Variable Measurement and Research Design

There are 81,505 NYSE and AMEX firm-quarter observations on the Compustat Quarterly Research and Active Files for the sample period 1989-2001. We limit ourselves to the post 1988 period as our key variables are derived from statement of cash flow data. NASDAQ firms are excluded from our analysis for the purpose of comparing our results to prior research (e.g., Collins and Hribar 2000). Elimination of firms with insufficient data to compute accruals and SUE reduces the sample to 62,972 firm-quarters. Further elimination of firms with insufficient data to compute share repurchases, insider trading, future stock returns and additional explanatory variables for our expectation models

produces the final sample of 58,030 firm-quarters. For our tests examining repurchase and insider trading in the following 6 months (i.e., POST period) the sample drops to 51,437 firm-quarters.

Insider trading data is recorded by Thomson Financial as missing when insiders do not trade in a period. To minimize data deletions, we assign a purchase ratio to observations with missing trading data based on the mean insider purchase ratio for a portfolio of similar-sized firms in the same calendar quarter (we match by size quintiles ranked on assets).⁷ We interpret this mean insider purchase ratio as the ratio that provides no information about insiders' valuation beliefs. Our results are robust to assigning the unconditional mean value of the insider purchase ratio to the firms with missing trade data, as well as to using the restricted sample with non-missing insider trading data.

Our research design is based on measures of operating accruals and SUE using quarterly data. Earnings are generally announced a few weeks after the end of the fiscal quarter, with financial statement information necessary to compute accruals being disclosed a few weeks after the earnings announcement. To ensure that investors have sufficient information about earnings and accruals, we begin the return measurement period for the accruals anomaly and SUE two months after the end of the fiscal quarter. In other words, for every quarter t that ends with month m , we rank firms according to the accruals and SUE anomalies, and we measure abnormal returns to these anomalies beginning at the start of month $m+3$ relative to the end of quarter t .⁸

⁷ Specifically, for the five size quintiles (LOW through to HIGH), the “no information” insider trading value is 0.60, 0.52, 0.45, 0.38 and 0.28 respectively.

⁸ Our choice to begin compounding six-month returns two months after the quarter-end is consistent with that of Collins and Hribar, who compound starting 18 trading days after the earnings announcement. Our compounding strategy will start about 42 trading days after the quarter-end, where theirs will typically start roughly 48 trading days after the quarter-end (assuming that earnings announcements are usually made about 30 trading days after the quarter (Easton and Zmijewski, 1993)).

We examine repurchase and insider trading activity in the 12-month period straddling the anomaly measurement period (i.e., from the start of month $m-3$ through to the end of month $m+8$ relative to the end of quarter t). We label the six-month period leading up to the abnormal return window (i.e., the six months ending at month $m+2$) as the “PRE” period. Likewise, we label the six-month period at the beginning of the abnormal return window (i.e., from the start of month $m+3$ to the end of month $m+8$) as the “POST” period. Figure 1 depicts the variable measurement windows for our tests. We examine both the PRE and POST periods because it is not clear when management will trade on information that is correlated with the anomalies. For example, the accruals anomaly suggests that firms reporting low levels of accruals experience improved stock returns going forward, and these returns persist for the duration of the following fiscal period (with some concentration around subsequent earnings announcement dates). A manager seeking to take advantage of mis-pricing that is correlated with extreme accruals periods may trade toward the end of an extreme accruals period or in the following fiscal periods as these abnormal returns start to be realized. Similar arguments can be made for the SUE anomaly. We note that our examination of trading during the PRE period assumes that managers have private information about firm valuation and possibly accounting information that allows managers to perceive mis-pricings prior to the return measurement periods for the anomalies. However, regulation on insider trading activity prohibits managers from trading in periods prior to major news events, including earnings announcements, which will limit manager’s ability to take advantage of the mispricing during the PRE period. If this is case, then the POST period may be considered a more powerful setting.

Insert Figure 1 here

We measure operating accruals and SUE as per Collins and Hribar (2000). We compute accruals as the difference between earnings and cash flows from operations using data from the statement of cash flow,

$$\text{Accruals}_t = \text{Earnings}_t - \text{CFO}_t \quad (4)$$

where Earnings_t is earnings from continuing operations (data #8) in quarter t and CFO_t is cash flow from operations (data #108) in quarter t . Accruals is then deflated by the average book value of total assets (data #44) in quarter t .

We compute SUE as the difference between net income in quarter t and net income in quarter $t-4$ deflated by market value of equity in quarter $t-4$,⁹

$$\text{SUE}_t = (\text{NI}_t - \text{NI}_{t-4}) / \text{MV_Equity}_{t-4} \quad (5)$$

where SUE_t is the standardized unexpected earnings in quarter t , NI_t is net income in quarter t (data #69), and MV_Equity_{t-4} is market value of equity computed as the closing share price (data #14) times the number of shares outstanding (data #61) at the end of quarter $t-4$.

For each anomaly, we compute size-adjusted returns during the future return

accumulation period by subtracting the value-weighted average return for all firms in the same size-matched decile, where size is measured as the market capitalization at the beginning of the fiscal period. Returns are calculated for a six-month period beginning two months after the end of the fiscal quarter. If a security delists, we calculate the return by reinvesting the remaining proceeds in the CRSP size-matched index until the end of the period. Consistent with previous research, we expect that firms with low (high) accruals and high (low) SUE will exhibit high (low) returns in the post-formation period.

Our research design creates over-lapping observations. Specifically, because we examine accruals and SUE in every quarter t along with repurchase and insider trading activity for the two-quarter PRE and POST periods, there is a one-quarter over-lap in our PRE and POST dependent variables. We correct for this autocorrelation using Huber-White robust standard errors, which are a generalization of the White (1980) standard errors that are robust to both serial correlation and heteroskedasticity (Rogers, 1993).

Table 1 presents descriptive statistics of the variables. All variables are winsorized at the 1% and 99% levels (i.e., for each variable we re-assign its value if it is less (greater) than the 1st (99th) percentile to the value of the 1st (99th) percentile). The mean (median) value for share repurchases is -0.17% (-0.09%) and untabulated statistics show 26.11% (15,149 observations) of our repurchase observations are positive (a positive value indicates that the firm repurchased shares in the two quarters measured). The mean (median) purchase ratio for insider traders is 43.15% (45.33%) and untabulated results document that for 19.45 % (11,287 observations) of firms the insiders are only sellers (insider's purchase ratio equals zero), whereas for 11.93 % (6,091 observations)

⁹ Bernard and Thomas (1989) and Collins and Hribar (2000) scale unexpected earnings by the standard deviation of the unexpected earnings. However, Bernard and Thomas (1990) show that the two approaches

the insiders are only buyers (insider's purchase ratio equal one). The mean (median) cash to total asset ratio is 0.08 (0.04) and the mean (median) debt to asset ratio is 0.27 (0.26). Firms have mean (median) capital expenditures of 0.02 (0.01) of total assets and a mean (median) dividend yield per quarter of 0.01 (0.00).

Insert Table 1 here

Table 2 presents Pearson and Spearman correlations for the variables in the analysis. We observe that net share repurchases are positively correlated with size and dividend yield. The latter supports the argument that repurchases and dividends are complements (Grullon and Michaely, 2002). As expected, net share repurchases are negatively correlated with debt and capital expenditures. Finally, we observe that the insider trading measure is negatively correlated with size, supporting previous findings that managers of large firms are more likely to sell shares.

Insert Table 2 here

4. Results

4.1. Expectation Model for Share Repurchases and for Insider Trading

Table 3 presents the results of our expectation models for share repurchases and for insider trading.¹⁰ The models include fiscal quarter, year, and industry dummies (not

provide similar results.

¹⁰ As discussed in Section 3.1, our repurchase measure is censored at -3.85% to avoid the influence of secondary equity issuances. We note however that we obtain similar results if we use an uncensored

tabulated). Column I presents the results of our OLS model of share repurchases. All explanatory variables, except cash, are statistically significant in the predicted direction consistent with previous empirical research (e.g., Dittmar, 2000).¹¹ There is considerable serial correlation in stock repurchases at the firm level, which is consistent with firms entering the open market to buy back shares over a number of fiscal periods. Large firms repurchase more shares in accordance with Dittmar's (2000) findings and in contrast to the asymmetric information hypothesis. Firms with high leverage and high levels of capital expenditures repurchase fewer shares, and firms that pay high dividend yields repurchase more shares suggesting that dividends and share repurchases are complements in our sample. Column II presents the results of our OLS model for insider trading. As predicted, managers from large firms sell more shares than managers of small firms, and insider trading is serially correlated. We use the residuals from the regressions in Table 3 as our measures of *Ab. Repurchase* and *Ab. Insider Trading*.

Insert Table 3 here

4.2. Portfolio Analysis

We begin our analysis by replicating earlier findings on the abnormal returns that could be earned by making investments according to the strategies underlying the accruals and SUE anomalies. We assign firms into ten portfolios in each of the 52 sample quarters

measure of repurchases or if we censor the value of repurchase to zero to retain only decreases in shares outstanding (i.e., net repurchases only).

¹¹ If we censor negative values of repurchase to zero as an alternative computation of repurchases, then we find that cash is positively related to repurchase consistent with previous literature such as Dittmar (2000).

based on accruals and SUE.¹² We compute abnormal returns for each portfolio in each of the 52 quarters in the sample. Table 4 reports the average of the 52 six-month size-adjusted returns for each portfolio in both the PRE and POST periods, and for the hedge portfolio using the extreme portfolios. We form hedge portfolios taking corresponding long and short positions in firms from the extreme decile rankings. For accruals, the hedge portfolio takes a long position in firms reporting low accruals and a short position in the high accruals firms. For the SUE anomaly, the hedge strategy takes a long position in the high SUE firms and a short position in the low SUE firms. We report a t-statistic, based on the average and standard deviation of the 52 six-month returns, which tests whether the hedge portfolio return is statistically different from zero.

The second column (i.e., labeled “Abnormal Returns POST”) of panel A replicates findings for the accruals anomaly. Consistent with earlier studies, the hedge portfolio generates an abnormal return of 4.27% ($t=4.63$) in the six months after the start of the portfolio formation period. These results are smaller in magnitude but comparable to the 5.56% abnormal return spread found by Collins and Hribar (2000) for the shorter 1988 to 1997.

The second column of panel B replicates previous findings for the SUE anomaly. The hedge portfolio generates an abnormal return of 5.97% ($t = 6.02$) in the six months following the portfolio formation period. This result is very close to Collins and Hribar’s (2000) results of average abnormal returns of 6.88%.

Insert Table 4 here

¹² We use data from 1989 to 2001, so we have 52 overlapping six-month periods (ending every fiscal quarter) for each firm.

In the next three sub-sections, we examine whether abnormal share repurchase volume and abnormal insider trading patterns are consistent with managers attempting to benefit from the mispricing predicted by the accruals and the SUE anomaly. For each anomaly, we examine abnormal share repurchases and abnormal insider trading during the six-month period concurrent with the anomaly formation period (PRE period), and during the six-month period following the anomaly formation period (POST period). Using raw share repurchases and raw insider trading does not affect qualitatively the results.

4.2.1. Accruals Anomaly

The last four columns in Table 4, panel A show a greater abnormal volume of share repurchases and a higher proportion of insider buy trades for firms reporting low levels of accruals compared to firms reporting high levels of accruals. These findings hold for repurchases and insider trading during the six-month period concurrent with the anomaly portfolio formation period (PRE) and the following six-month period (POST).

These findings suggest that managers' repurchase and personal trading decisions are consistent with the trading strategy underlying the accruals anomaly. Our results with insider trading corroborate Beneish and Vargus' (2002) findings that insider trading and accruals are correlated. The strong pattern between repurchase activity and operating accruals is also related to the external financing measure developed in Richardson and Sloan (2003). Specifically, our finding of a contemporaneous correlation between repurchase activity in the PRE period and operating accruals is related to the link established in Richardson and Sloan between aggregate external financing and change in

net operating assets (a broader measure of accruals).

4.2.2. SUE Anomaly

In panel B of Table 4, we do not observe that managers adjust the volume of share repurchases or insider trades according to the SUE anomaly. Moreover, we document abnormal repurchase volume and insider trading in the direction opposite to what would be predicted by the SUE anomaly. Overall, the SUE results suggest that share repurchase volume and managerial trading behavior are not consistent with the mispricing predicted by the SUE anomaly. We discuss the possibility that this latter result is due to portfolio rebalancing by insiders in the next section.

4.3 Regression Analysis

Table 5 presents multiple regression results that capture correlation across the accruals and the SUE anomalies.¹³ We include two transformations of the anomaly variables. First, we use the anomalies' *ranks* (coded as a discrete variable taking values from 1 to 10) as an explanatory variable. This variable assumes a linear relation between the returns generated by the anomalies and the decile rankings. Because previous findings document that the anomaly returns are stronger in the extreme portfolios, we also define a *dummy* variable coded '-1' if the firm belongs to the first decile (low accruals and low SUE), '0' if the firm belongs to the intermediate portfolios (deciles 2 to 9) and '1' if the firm is included in the highest decile (high accruals and high SUE). The Table 5 regressions include the same control variables employed in the repurchase and insider trading expectation models reported in Table 3.

We also include *Past_returns* in the model. The rationale behind this variable is that the SUE anomaly predicts that insiders should sell less stock and repurchase less

shares following positive earnings surprises. However, it can be argued that insiders will rebalance their equity portfolios by selling own firm stock after positive earnings surprises because they are normally associated with stock price run-ups (see evidence in Table 4 – Panel B). This behavior potentially confounds our interpretation of the SUE tests.

Panel A of table 5 presents the results of six regression specifications for share repurchases. In Columns I to III, both the anomalies variables and share repurchases are measured during the two quarters of the PRE period. As in Table 4, we also include results (Columns IV to VI), in which we measure share repurchases during the two quarters of the POST period.¹⁴

Consistent with the portfolio analysis in Table 4, the estimated coefficients for *Accruals_ranks* are negative and significant for share repurchases in both the PRE and POST periods. The coefficients on the *Accruals_dummy* are also negative in both the PRE and POST periods, and robust to the inclusion of *Past_returns* in the model. We find in panel B of Table 5 a consistently negative relation between insider trading and accruals in the PRE but no relation between insider trading and the accruals variables in the POST period. This later result may be due to the conservative estimate of the POST period. Because information in accruals is available to managers before two months after the end of the quarter, managers do not need to wait two months to trade on the accruals information. Overall, the results suggest that marginal trading decisions of management

¹³ Our inference is unaffected if we instead run separate regressions for each anomaly.

¹⁴ In unreported tests we change our PRE and POST intervals to match the fiscal quarter. Specifically, the PRE period ends at the end of the fiscal quarter and the POST period starts in the beginning of the following quarter. Using these alternate windows we find consistent results with the accruals anomaly for both share repurchases and insider trading in both periods. For the SUE anomaly, the results are similar to the ones presented in the paper and inconsistent with the mispricing explanation.

(both repurchases and insider trading) are consistent with the mispricing explanation for accruals.

The estimated coefficients on the *SUE* variables in Panels A of Table 5 are opposite the predicted sign in both the PRE and the POST periods (with the exception of the *SUE_dummy* in the POST period). For the insider trading regressions (Panel B), the coefficients on the *SUE* variables are negative and statistically significant. These findings corroborate the portfolio results in Table 4 that suggest that share repurchase volume and insider trading are not consistent with the mispricing predicted by the *SUE* anomaly.

Overall, the regression results corroborate the portfolio analysis suggesting that managers' repurchase and insider trading behavior varies consistently with the information underlying the operating accruals trading strategy. On the other hand, we do not find corroborative evidence for the *SUE* anomaly.¹⁵

Insert Table 5 here

5. Summary and Conclusions

In this paper, we examine whether managers' trading decisions (both at a firm and personal level) are correlated with trading strategies suggested by the operating accruals and the post-earnings announcement drift (*SUE*) anomalies. These two anomalies are especially suited for the purposes of this paper because managers are uniquely informed about financial reporting and are in the best position to observe pricing deviations from

¹⁵ Firms without share repurchase programs may find it too costly to start a repurchase program simply to take advantage of transient mispricing. Therefore, we repeat our analysis using 43,391 firms with some history of repurchasing shares. For this reduced sample we find results very similar to those reported in the paper.

fundamental value.

The use of managerial trading activity to infer managers' private valuation about their own securities is a unique feature of our research design and we discuss advantage and disadvantage of this methodology in the paper. We argue that if managers are skilled at identifying equity mispricing and an anomaly accurately captures situations where stock prices deviate from firms' fundamental value, then managers' marginal trading decisions through repurchase programs and on personal accounts will be correlated with the anomaly trading strategy. On the other hand, restrictions on the managers' ability to take advantage of temporary mispricing due to insider trading restrictions, prohibitive trading costs, or risk associated with trading strategy may render the interpretation of null results inconclusive.

We provide corroborative evidence for the accruals anomaly, i.e., managers' repurchase and insider trading behavior varies consistently with the information underlying the operating accruals trading strategy. Low (high) accruals firms repurchase more (less) shares, and managers of low (high) accruals firms buy more (less) shares on their personal accounts. On the other hand, we do not find corroborative evidence for the SUE anomaly. The fact that we find corroborating evidence for the accruals anomaly but not the SUE anomaly suggests that under the assumption that the SUE anomaly truly captures mispricing, trading on information of the type embedded in SUE is more costly to managers than trading on information of the type embedded in accruals. Furthermore, given that our research design has sufficient power to corroborate an anomaly that relies on financial statement information (i.e., accruals), future research could exploit this design to examine the validity of other financial-statement-based "anomalies".

References

- Ang, J. S., and Cheng, Y., 2003. Direct evidence on the market-driven acquisitions theory. Working paper, Florida State University.
- Baker, M. and Wurgler, J., 2002. Market Timing and Capital Structure, *Journal of Finance* 57 (1), 1-32.
- Ball, R., 1992. The earnings–price anomaly. *Journal of Accounting and Economics* 15, 319–345.
- Ball, R., and Bartov, E., 1996. How naive is the stock market’s use of earnings information? *Journal of Accounting and Economics* 21, 319–337.
- Ball, R., and Brown, P., 1968. An empirical evaluation of accounting income numbers. *Journal of Accounting Research* 6, 159–177.
- Beneish, M., and Vargus, M., 2002. Insider Trading, Earnings Quality, and Accruals Mispricing. *The Accounting Review*, 77 (4), 755-791.
- Bernard, V., and Thomas, J., 1989. Post-earnings-announcement drift: delayed price response or risk premium? *Journal of Accounting Research* 27, 1–48.
- Bernard, V., and Thomas, J., 1990. Evidence that stock prices do not fully reflect the implications of current earnings for future earnings. *Journal of Accounting and Economics* 13, 305–340.
- Brav, A., Graham, J., Harvey, C., and Michaely, R., 2005. Payout Policy in the 21st Century. Forthcoming *Journal of Financial Economics*.
- Bushee, B., and Raedy, J., 2003. Factors Affecting the Implementability of Stock Market Trading Strategies. Working paper, University of Pennsylvania.
- Chan, K., Jegadeesh, N., and Lakonishok, J., 1996. Momentum strategies. *Journal of Finance* 51, 1681–1713.
- Collins, D., and Hribar, P., 2000. Earnings-based and accrual-based market anomalies: one effect or two? *Journal of Accounting and Economics* 29, 101–123.
- D’Mello, R., and Shroff, P., 2000. Equity Undervaluation and Decisions Related to Repurchase Tender Offers: An Empirical Investigation. *Journal of Finance*, 55 (5), 2399-2424.
- Dittmar, A., 2000. Why Do Firms Repurchase Stock? *Journal of Business* 73 (3), 331-355.

- Dong, M., Hirshleifer, D., Richardson S., and Teoh S., 2005. Does Investor Misvaluation Drive the Takeover Market? Forthcoming, *Journal of Finance*.
- Easton, P., and Zmijewski, M., 1993. SEC Form 10K/10Q Reports and Annual Reports to Shareholders: Reporting Lag and Squared Market Model Prediction Errors. *Journal of Accounting Research* 31 (1), 113-129.
- Fama, E., 1998. Market efficiency, long-term returns, and behavioral finance. *Journal of Financial Economics* 49, 283–306.
- Foster, G., Olsen, C., and Shevlin, T., 1984. Earnings releases, anomalies and the behavior of security returns. *The Accounting Review* 59, 574–603.
- Grullon, G., and Michaely, R., 2002. Dividends, Share repurchases, and the Substitution Hypothesis. *Journal of Finance* 57 (4), 1649-1684.
- Ikenberry, D., Lakonishok, J., and Vermaelen, T., 1995. Market underreaction to open market share repurchases. *Journal of Financial Economics* 39, 181-208.
- Ikenberry, D., Lakonishok, J., and Vermaelen, T., 2000. Stock Repurchases in Canada: Performance and Strategic Trading. *Journal of Finance* 55 (5), 2373-2397.
- Jones, C., and Litzenberger, R., 1970. Quarterly earnings reports and intermediate stock price trends. *Journal of Finance* 25, 143–148.
- Kothari, S., 2001. Capital Markets Research in Accounting, *Journal of Accounting and Economics* 31, 105-231.
- Lee, C., 2001. Market Efficiency and Accounting Research: a Discussion of ‘Capital Markets Research in Accounting’ by S. P. Kothari, *Journal of Accounting and Economics* 31, 233-253.
- Pincus, M., Rajgopal, S., and Venkatachalam, M., 2003. The Accrual Anomaly: International Evidence. Working paper, University of Iowa.
- Piotroski, J., and Roulstone, D., 2005. Do Insider Trades Reflect Superior Knowledge about Future Cash Flow Realizations? *Journal of Accounting and Economics* 39(1), 55-81
- Polk, C. and Sapienza, P., 2004. The Real Effects of Investor Sentiment, working paper, Northwestern University.
- Rhodes-Kropf, M., Robinson D., and Viswanathan S., 2003. Valuation Waves and Merger Activity: The Empirical Evidence. Working paper, Columbia University.
- Richardson, S., Sloan, R., Soliman, M., and Tuna, I., 2005. Accrual Reliability, Earnings

- Persistence and Stock Prices. Forthcoming, *Journal of Accounting and Economics*.
- Richardson, S., and Sloan R., 2003. External financing, Capital Investment and Future Stock Returns. Working paper, University of Pennsylvania.
- Rogers, W., 1993. Regression standard errors in clustered samples. *Stata Technical Bulletin Reprints*, vol. 3., College Station, Texas: Stata Press, 83-94.
- Rozeff, M., and Zaman, M., 1988. Market Efficiency and Insider Trading: New Evidence. *Journal of Business*, 61 (1).
- Rozeff, M., and Zaman, M., 1998. Overreaction and Insider Trading: Evidence from Growth and Value Portfolios. *Journal of Finance* 53 (2), 701-716.
- Schwert, G., 2003. Anomalies and Market Efficiency. Chapter 15, *Handbook of the Economics of Finance*, eds. George Constantinides, Milton Harris, and René Stulz, North-Holland, pp. 937-972.
- Seyhun, H., 1986. Insider Profits, Costs of Trading, and Market Efficiency. *Journal of Financial Economics* 16 (June), 189-212.
- Seyhun, H., 1992. Why Does Aggregate Insider Trading Predict Future Stock Returns, *The Quarterly Journal of Economics* 107 (4), 1303-1331.
- Sloan, R., 1996. Do stock prices fully reflect information in accruals and cash flows about future earnings? *The Accounting Review* 71, 289–316.
- Stephens, C., and Weisbach, M., 1998. Actual share reacquisitions in open market repurchase programs. *Journal of Finance* 53, 313-334.
- White, H., 1980. A Heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica* 48, 817-838.

Table 1 –Descriptive Statistics

Variables	N	Mean	STD	Min	Median	Max
<i>Repurchase (%)</i>	58,030	-0.17	2.35	-3.85	-0.09	11.43
<i>Ins. Trading (%)</i>	58,030	43.15	29.33	0.00	45.33	100.00
<i>Size</i>	58,030	6.30	1.84	2.04	6.30	10.53
<i>Cash</i>	58,030	0.08	0.12	0.00	0.04	0.64
<i>Debt</i>	58,030	0.27	0.19	0.00	0.26	0.89
<i>Cap. Expenses</i>	58,030	0.02	0.02	0.00	0.01	0.11
<i>Dividend Yield</i>	58,030	0.01	0.01	0.00	0.00	0.05

Repurchase is the percentage of shares repurchased in the six-month period ending two months after quarter t. We calculate *Repurchase* as -1 multiplied by the percentage change in shares outstanding as reported by CRSP. Thus, the measure is increasing in the number of shares repurchases. We winsorize *Repurchase* to be no less than -3.85%, i.e., if the actual repurchase value is less than -3.85% of shares outstanding, then we replace it with -3.85%. This number represents the 1st percentile in our sample of the distribution of a two-quarter change in treasury stock from Compustat (data #98) deflated by the market value of total shares outstanding at the start of the period.

Ins. Trading is the insider purchase ratio measured as the ratio of the number of shares purchased by insiders in the six-month period ending two months after quarter t, scaled by total number of shares traded (purchases plus sales) by insiders during the same period. Insider trading is limited to open market transactions of the top five executives coded as "CB", "CEO", "CFO", "CO", and "P" by Thomson Financial First Call Insiders Data.

Size is the log of total assets (data #44) measured at the end of the quarter t-2.

Cash is the ratio of cash and cash equivalents (data #36) to total assets (data #44) at the end of the quarter t-2.

Debt is the ratio of long-term debt (data #51) plus the debt included in current liabilities (data #45) to total assets (data #44) at the end of the quarter t-2.

Cap. Expenses is the ratio of capital expenditures (data #90) to the average total assets (data #44). This variable is measured at the end of the quarter t-2.

Dividend Yield is computed as the dividend-per-share (data #16) divided by the mean closing price in the quarter ((data #12 + data #13 + data #14)/3). It is calculated for the six month period ending in quarter t-2.

Table 2 – Correlation Matrix

Variables	<i>Repurchase (%)</i>	<i>Ins. Trading (%)</i>	<i>Size</i>	<i>Cash</i>	<i>Debt</i>	<i>Cap. Expenses</i>	<i>Dividend Yield</i>
<i>Repurchase (%)</i>	---	0.05 <i><.0001</i>	0.06 <i><.0001</i>	-0.01 <i>0.129</i>	-0.06 <i><.0001</i>	-0.05 <i><.0001</i>	0.08 <i><.0001</i>
<i>Ins. Trading (%)</i>	0.07 <i><.0001</i>	---	-0.37 <i><.0001</i>	0.04 <i><.0001</i>	0.04 <i><.0001</i>	-0.06 <i><.0001</i>	-0.03 <i><.0001</i>
<i>Size</i>	0.04 <i><.0001</i>	-0.55 <i><.0001</i>	---	-0.29 <i><.0001</i>	0.17 <i><.0001</i>	0.06 <i><.0001</i>	0.23 <i><.0001</i>
<i>Cash</i>	0.00 <i>0.87</i>	0.06 <i><.0001</i>	-0.21 <i><.0001</i>	---	-0.37 <i><.0001</i>	-0.08 <i><.0001</i>	-0.11 <i><.0001</i>
<i>Debt</i>	-0.07 <i><.0001</i>	-0.03 <i><.0001</i>	0.20 <i><.0001</i>	-0.42 <i><.0001</i>	---	0.01 <i>0.17</i>	-0.02 <i><.0001</i>
<i>Cap. Expenses</i>	-0.04 <i><.0001</i>	-0.15 <i><.0001</i>	0.18 <i><.0001</i>	-0.08 <i><.0001</i>	-0.01 <i>0.03</i>	---	-0.05 <i><.0001</i>
<i>Dividend Yield</i>	0.16 <i><.0001</i>	-0.16 <i><.0001</i>	0.35 <i><.0001</i>	-0.14 <i><.0001</i>	-0.03 <i><.0001</i>	0.07 <i><.0001</i>	---

Pairwise Correlation Matrix – Pearson correlations are reported above the diagonal and Spearman correlations below the diagonal. P-values are shown in italics below correlations.

Table 2 – Correlation Matrix (cont'd)

Repurchase is the percentage of shares repurchased in the six-month period ending two months after quarter t . We calculate *Repurchase* as -1 multiplied by the percentage change in shares outstanding as reported by CRSP. Thus, the measure is increasing in the number of shares repurchases. We winsorize *Repurchase* to be no less than -3.85%, i.e., if the actual repurchase value is less than -3.85% of shares outstanding, then we replace it with -3.85%. This number represents the 1st percentile in our sample of the distribution of a two-quarter change in treasury stock from Compustat (data #98) deflated by the market value of total shares outstanding at the start of the period.

Ins. Trading is the insider purchase ratio measured as the ratio of the number of shares purchased by insiders in the six-month period ending two months after quarter t , scaled by total number of shares traded (purchases plus sales) by insiders during the same period. Insider trading is limited to open market transactions of the top five executives coded as "CB", "CEO", "CFO", "CO", and "P" by Thomson Financial First Call Insiders Data.

Size is the log of total assets (data #44) measured at the end of the quarter $t-2$.

Cash is the ratio of cash and cash equivalents (data #36) to total assets (data #44) at the end of the quarter $t-2$.

Debt is the ratio of long-term debt (data #51) plus the debt included in current liabilities (data #45) to total assets (data #44) at the end of the quarter $t-2$.

Cap. Expenses is the ratio of capital expenditures (data #90) to the average total assets (data #44). This variable is measured at the end of the quarter $t-2$.

Dividend Yield is computed as the dividend-per-share (data #16) divided by the mean closing price in the quarter $((\text{data \#12} + \text{data \#13} + \text{data \#14})/3)$. It is calculated for the six month period ending in quarter $t-2$.

Table 3 – Expectation Model for *Repurchase* and *Insider Trading*

Indep. Variables	<i>Repurchase</i> (%)		<i>Ins. Trading</i> (%)	
	Predicted Sign	I	Predicted Sign	II
<i>Intercept</i>		-0.33 (-0.91)		60.83*** (13.78)
<i>Repurchase</i> _{<i>t</i>-2}	+	0.15*** (18.87)		
<i>Ins. Trading</i> _{<i>t</i>-2}	+			0.32*** (40.83)
<i>Size</i>	+/-	0.05*** (4.64)	-	-4.21*** (-34.66)
<i>Cash</i>	+	0.02 (0.10)		
<i>Debt</i>	-	-0.81*** (-7.88)		
<i>Cap. Expenses</i>	-	-5.07*** (-5.92)		
<i>Dividend Yield</i>	+/-	12.02*** (6.20)		
R-square (%)		6.69		24.02
Observations		58,030		58,030

Column I presents the results of an OLS model examining the determinants of *Repurchase*. Column II presents the results of an OLS model examining the determinants of *Ins. Trading*. Firm-quarter observations are drawn from 1989 to 2001. Fiscal quarters, year and industry dummies are included in the model but not tabulated in the results. t-statistics are presented in parenthesis below coefficient estimates based on Huber-White robust standard errors. *, **, and *** indicate two-tailed statistical significance at 10, 5, and 1 percent levels, respectively.

Table 3 – Expectation Model for *Repurchase* and *Insider Trading* (cont'd)

Repurchase is the percentage of shares repurchased in the six-month period ending two months after quarter t . We calculate *Repurchase* as -1 multiplied by the percentage change in shares outstanding as reported by CRSP. Thus, the measure is increasing in the number of shares repurchases. We winsorize *Repurchase* to be no less than -3.85%, i.e., if the actual repurchase value is less than -3.85% of shares outstanding, then we replace it with -3.85%. This number represents the 1st percentile in our sample of the distribution of a two-quarter change in treasury stock from Compustat (data #98) deflated by the market value of total shares outstanding at the start of the period.

Ins. Trading is the insider purchase ratio measured as the ratio of the number of shares purchased by insiders in the six-month period ending two months after quarter t , scaled by total number of shares traded (purchases plus sales) by insiders during the same period. Insider trading is limited to open market transactions of the top five executives coded as "CB", "CEO", "CFO", "CO", and "P" by Thomson Financial First Call Insiders Data.

Size is the log of total assets (data #44) measured at the end of the quarter $t-2$.

Cash is the ratio of cash and cash equivalents (data #36) to total assets (data #44) at the end of the quarter $t-2$.

Debt is the ratio of long-term debt (data #51) plus the debt included in current liabilities (data #45) to total assets (data #44) at the end of the quarter $t-2$.

Cap. Expenses is the ratio of capital expenditures (data #90) to the average total assets (data #44). This variable is measured at the end of the quarter $t-2$.

Dividend Yield is computed as the dividend-per-share (data #16) divided by the mean closing price in the quarter $((\text{data \#12} + \text{data \#13} + \text{data \#14})/3)$. It is calculated for the six month period ending in quarter $t-2$.

Table 4 - Portfolio Analysis

Panel A – Portfolio Strategy for the Accruals Anomaly

<i>Accruals Ranks</i>	<i>Obs</i>	<i>Ab. Returns (%)</i>		<i>Ab. Repurchase (%)</i>		<i>Ab. Ins. Trading (%)</i>	
		<i>Pre</i>	<i>Post</i>	<i>Pre</i>	<i>Post</i>	<i>Pre</i>	<i>Post</i>
1 (Low)	5,779	-1.34	4.32	0.07	0.04	0.93	-0.27
2	5,811	0.85	3.15	0.07	0.06	0.17	0.09
3	5,808	0.82	1.91	0.09	0.06	-0.02	0.34
4	5,805	0.89	2.14	-0.02	0.07	0.57	0.13
5	5,799	0.52	1.48	0.03	0.05	0.39	0.28
6	5,819	1.11	1.39	0.00	-0.04	0.07	-0.25
7	5,808	1.60	1.20	-0.06	0.00	-0.13	0.67
8	5,805	1.38	0.65	-0.01	-0.04	-0.70	-0.40
9	5,814	2.17	-0.03	-0.01	0.00	-0.25	0.01
10 (High)	5,782	2.75	0.05	-0.15	-0.08	-1.01	-1.10
Hedge: 1-10		-4.09	4.27	0.22	0.12	1.94	0.84
t-test		-3.83	4.63	5.46	2.82	3.75	2.14
Pred. Sign			+	+	+	+	+
Obs		58,030	58,030	58,030	51,437	58,030	51,437

Panel B – Portfolio Strategy for the SUE Anomaly

<i>SUE Ranks</i>	<i>Obs</i>	<i>Ab. Returns (%)</i>		<i>Ab. Repurchase (%)</i>		<i>Ab. Ins. Trading (%)</i>	
		<i>Pre</i>	<i>Post</i>	<i>Pre</i>	<i>Post</i>	<i>Pre</i>	<i>Post</i>
1 (Low)	5,779	-10.33	1.05	0.04	-0.01	4.13	4.17
2	5,811	-8.51	-0.44	0.17	0.19	3.41	2.66
3	5,808	-7.29	-0.69	0.23	0.13	2.71	1.72
4	5,805	-4.60	-0.14	0.24	0.20	0.52	0.08
5	5,799	-1.37	0.56	0.13	0.18	-0.76	-1.56
6	5,819	2.73	1.04	-0.01	-0.02	-2.93	-2.68
7	5,808	5.51	1.82	-0.17	-0.09	-3.21	-2.53
8	5,805	7.81	3.10	-0.20	-0.15	-2.60	-1.75
9	5,814	10.34	2.94	-0.25	-0.16	-1.59	-0.69
10 (High)	5,782	16.45	7.03	-0.17	-0.14	0.37	0.33
Hedge: 10-1		26.78	5.97	-0.22	-0.14	-3.76	-3.84
t-test		21.24	6.02	-4.38	-2.85	-6.69	-6.40
Pred. Sign			+	+	+	+	+
Obs		58,030	58,030	58,030	51,437	58,030	51,437

Table 4 - Portfolio Analysis (cont'd)

Stock returns are measured using compounded buy-hold returns, inclusive of dividends and other distributions. *Ab. Returns* t is the size-adjusted returns calculated by deducting the corresponding value-weighted return for all available firms in the same size-matched decile, where size is measured using market capitalization as of the beginning of the year. Returns are calculated for two six-month intervals. The PRE period begins at the start of month $m-3$ and ends with month $m+2$ (where month m is the end of fiscal quarter). The POST period begins two months after the end of the fiscal quarter t (i.e., from the start of month $m+3$ to $m+8$). For firms that delist during our future return window, we calculate the remaining return by reinvesting any remaining proceeds in the value-weighted market portfolio size-adjusted abnormal return ending two months after the end of the quarter t . *Ab. Repurchase* t is the abnormal volume of share repurchases estimated as the residual of the regression in Table 3 and *Ab. Ins. Trading* t is the abnormal volume of insider trading estimated as the residual of the regression in Table 3. Hedge 1-10 (10-1) represents the net return generated by taking a long position in the Low (High) portfolio and an equal sized short position in the High (Low) portfolio. *Pre* corresponds to the two-quarter period (quarters t and $t-1$) ending two months after the end of quarter t . *Post* corresponds to the two-quarter period (quarters $t+1$ and $t+2$) starting two months after the end of quarter t . The t -statistic tests whether hedge is statistically different from zero. *, **, and *** indicate two-tailed statistical significance at 10, 5, and 1 percent levels.

Repurchase is the percentage of shares repurchased in the six-month period ending two months after quarter t . We calculate *Repurchase* as -1 multiplied by the percentage change in shares outstanding as reported by CRSP. Thus, the measure is increasing in the number of shares repurchases. We winsorize *Repurchase* to be no less than -3.85%, i.e., if the actual repurchase value is less than -3.85% of shares outstanding, then we replace it with -3.85%. This number represents the 1st percentile in our sample of the distribution of a two-quarter change in treasury stock from Compustat (data #98) deflated by the market value of total shares outstanding at the start of the period.

Ins. Trading is the insider purchase ratio measured as the ratio of the number of shares purchased by insiders in the six-month period ending two months after quarter t , scaled by total number of shares traded (purchases plus sales) by insiders during the same period. Insider trading is limited to open market transactions of the top five executives coded as "CB", "CEO", "CFO", "CO", and "P" by Thomson Financial First Call Insiders Data.

Accruals are computed as the difference between earnings from continuing operations (data #8) in quarter t and cash flows from operations (data #108) in quarter t using data from the statement of cash flow divided by average total assets (data #44) at the end of the quarters $t-1$ and t .

SUE is computed as the difference between the net income (data #69) in the quarter t and the net income (data #69) in the quarter $t-4$ deflated by the market value of equity (data #14 * data #61) in the quarter $t-4$.

Table 5 – Multiple Regressions

Panel A –Analysis for Share repurchases

Ind. Variable	Pred. Sign	Repurchase - Pre-period			Repurchase - Post- period		
		I	II	III	IV	V	VI
<i>Intercept</i>		0.01 (0.02)	-0.33 (-0.92)	-0.26 (-0.72)	-0.15 (-0.38)	-0.38 (-0.95)	-0.33 (-0.83)
<i>Repurchase</i> _{t-2}	+	0.15*** (18.45)	0.15*** (18.81)	0.15*** (18.42)	0.16*** (17.51)	0.16*** (17.83)	0.16*** (17.43)
<i>Size</i>	+/-	0.05*** (4.46)	0.05*** (4.60)	0.05*** (4.59)	0.05*** (3.98)	0.05*** (3.98)	0.05*** (3.98)
<i>Cash</i>	+	0.00 0.01	0.02 0.12	0.06 0.32	0.10 0.45	0.10 0.46	0.14 0.64
<i>Debt</i>	-	-0.80*** (-7.82)	-0.82*** (-7.92)	-0.83*** (-8.01)	-0.87*** (-7.96)	-0.85*** (-7.81)	-0.86*** (-7.86)
<i>Cap. Expenses</i>	-	-5.77*** (-6.79)	-5.32*** (-6.20)	-5.31*** (-6.24)	-5.09*** (-5.56)	-4.95*** (-5.35)	-5.06*** (-5.51)
<i>Dividend Yield</i>	+/-	10.97*** (5.68)	11.85*** (6.10)	10.39*** (5.36)	11.31*** (5.56)	12.21*** (5.96)	10.73*** (5.25)
<i>Accruals_ranks</i>	-	-0.01*** (-3.68)			-0.01** (-2.47)		
<i>SUE_ranks</i>	+	-0.05*** (-12.86)			-0.04*** (-8.39)		
<i>Accruals_dummy</i>	-		-0.10*** (-4.33)	-0.08*** (-3.39)		-0.05* (-1.91)	-0.04* (-1.75)
<i>SUE_dummy</i>	+		-0.10*** (-4.23)	-0.07*** (-2.80)		-0.06** (-2.38)	0.01 (0.61)
<i>Past_returns</i>	-			-0.01*** (-17.66)			-0.01*** (-16.91)
R-square (%)		7.14	6.77	7.49	7.31	7.12	7.89
Observations		58,030	58,030	58,030	51,437	51,437	51,437

Table 5 – Multiple Regressions (cont'd)**Panel B – Analysis for Insider Trading**

Ind. Variable	Pred. Sign	Ins. Trading – Pre-period			Ins. Trading – Post-period		
		I	II	III	IV	V	VI
<i>Intercept</i>		64.93*** (14.74)	60.90*** (13.82)	61.97*** (13.96)	66.08*** (13.88)	63.32*** (13.29)	64.26*** (13.35)
<i>Ins. Trading</i> _{t-2}	+	0.32*** (40.60)	0.32*** (40.79)	0.31*** (39.52)	0.31*** (37.99)	0.32*** (38.44)	0.31*** (37.20)
<i>Size</i>	-	-4.27*** (-35.10)	-4.23*** (-34.77)	-4.29*** (-35.11)	-4.32*** (-33.67)	-4.29*** (-33.45)	-4.34*** (-33.77)
<i>Accruals_ranks</i>	-	-0.09** (-2.23)			0.01 (0.19)		
<i>SUE_ranks</i>	+	-0.66*** (-14.91)			-0.51*** (-11.00)		
<i>Accruals_dummy</i>	-		-0.76*** (-3.05)	-0.63** (-2.52)		-0.17 (-0.63)	-0.15 (-0.58)
<i>SUE_dummy</i>	+		-1.80*** (-7.06)	-1.56*** (-6.17)		-1.90*** (-6.92)	-1.44*** (-5.25)
<i>Past_returns</i>	-			-0.04*** (-10.29)			-0.04*** (-9.76)
R-square (%)		24.46	24.12	24.31	24.57	24.41	24.60
Observations		58,030	58,030	58,030	51,437	51,437	51,437

Panel A presents the results of an OLS model examining the determinants of *Repurchase* and the effects of the anomalies variables. Panel B presents the results of an OLS model examining the determinants of *Ins. Trading* and the effects of the anomalies variables. Firm-quarter observations are drawn from 1989 to 2001. *Pre-Period* corresponds to the two-quarter period (quarters t and t-1) ending two months after the end of quarter t. *Post-Period* corresponds to the two-quarter period (quarters t+1 and t+2) starting two months after the end of quarter t. Fiscal quarters, year and industry dummies are included in the model but not tabulated in the results. t-statistic are presented in parenthesis below coefficient estimates based on Huber-White robust standard errors. *, **, and *** indicate two-tailed statistical significance at 10, 5, and 1 percent levels.

Table 5 – Multiple Regressions (cont'd)

Repurchase is the percentage of shares repurchased in the six-month period ending two months after quarter *t*. We calculate *Repurchase* as -1 multiplied by the percentage change in shares outstanding as reported by CRSP. Thus, the measure is increasing in the number of shares repurchases. We winsorize *Repurchase* to be no less than -3.85%, i.e., if the actual repurchase value is less than -3.85% of shares outstanding, then we replace it with -3.85%. This number represents the 1st percentile in our sample of the distribution of a two-quarter change in treasury stock from Compustat (data #98) deflated by the market value of total shares outstanding at the start of the period.

Ins. Trading is the insider purchase ratio measured as the ratio of the number of shares purchased by insiders in the six-month period ending two months after quarter *t*, scaled by total number of shares traded (purchases plus sales) by insiders during the same period. Insider trading is limited to open market transactions of the top five executives coded as "CB", "CEO", "CFO", "CO", and "P" by Thomson Financial First Call Insiders Data.

Size is the log of total assets (data #44) measured at the end of the quarter *t-2*.

Cash is the ratio of cash and cash equivalents (data #36) to total assets (data #44) at the end of the quarter *t-2*.

Debt is the ratio of long-term debt (data #51) plus the debt included in current liabilities (data #45) to total assets (data #44) at the end of the quarter *t-2*.

Cap. Expenses is the ratio of capital expenditures (data #90) to the average total assets (data #44). This variable is measured at the end of the quarter *t-2*.

Dividend Yield is computed as the dividend-per-share (data #16) divided by the mean closing price in the quarter $((\text{data \#12} + \text{data \#13} + \text{data \#14})/3)$. It is calculated for the six month period ending in quarter *t-2*.

Accruals are computed as the difference between earnings from continuing operations (data #8) in quarter *t* and cash flows from operations (data #108) in quarter *t* using data from the statement of cash flow divided by average total assets (data #44) at the end of the quarters *t-1* and *t*.

SUE is computed as the difference between the net income (data #69) in the quarter *t* and the net income (data #69) in the quarter *t-4* deflated by the market value of equity (data #14 * data #61) in the quarter *t-4*.

Accruals_ranks is a discrete variable coded 1 to 10 depending on which portfolio the firm belongs to in the current quarter ranked by operating accruals at quarter *t*.

SUE_ranks: a discrete variable coded 1 to 10 depending on which portfolio the firm belongs to in the current quarter ranked by SUE at quarter *t*.

Accruals_dummy: a dummy variable coded '-1' if the firm belong to the first decile (low accruals), '0' if the firm belong to the intermediate portfolios (deciles 2 to 9) and '1' if the firm is included in the highest decile (high accruals) at quarter *t*.

SUE_dummy: a dummy variable coded '-1' if the firm belong to the first decile (low SUE), '0' if the firm belong to the intermediate portfolios (deciles 2 to 9) and '1' if the firm is included in the highest decile (high SUE) at quarter *t*.

Past_returns is the six-month period stock return ending at the end of the fiscal quarter *t*.

Figure 1 – Timeline of Variable Measurement

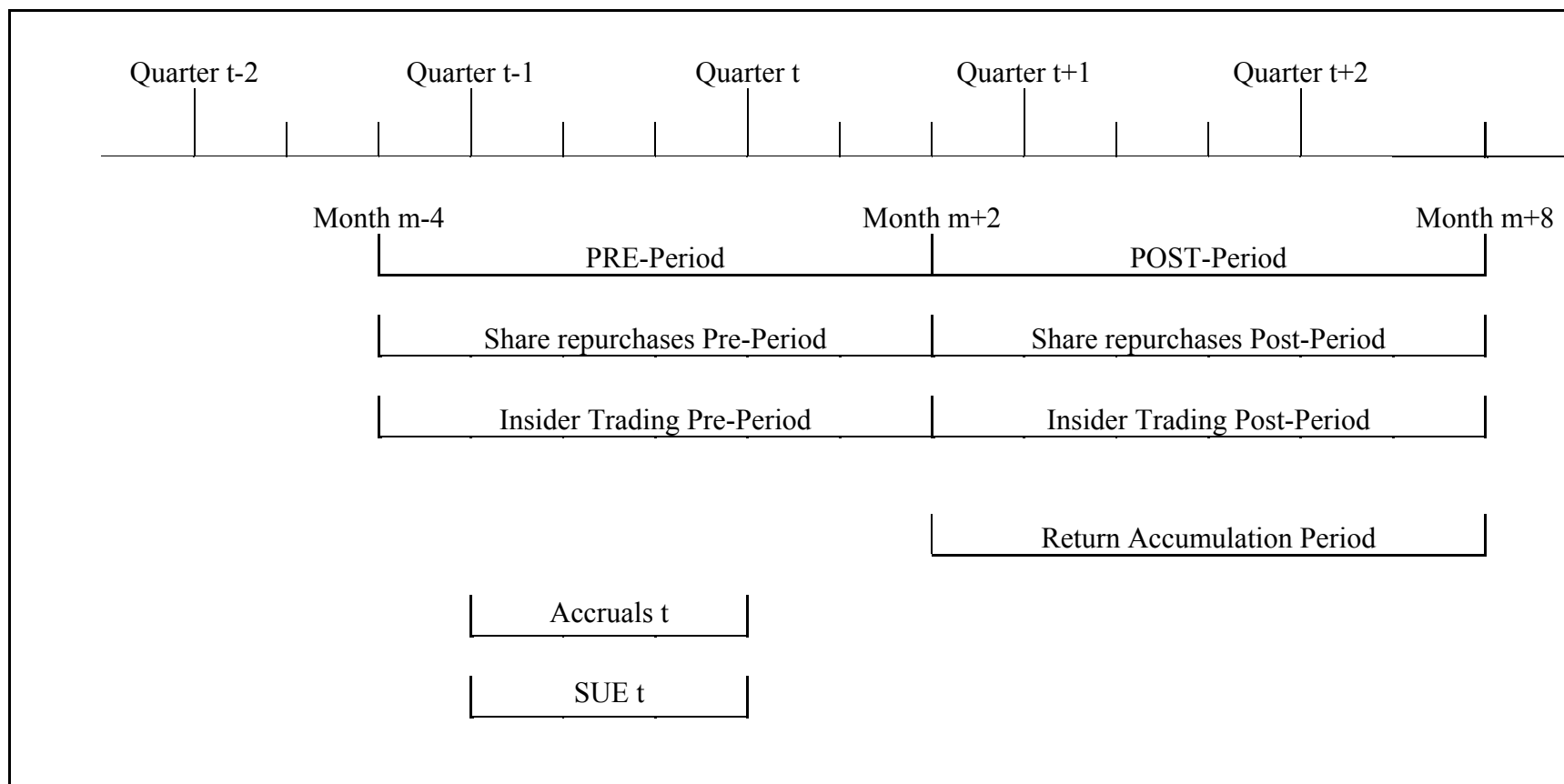


Figure 1 – Timeline of Variable Measurement (cont'd)

Repurchase is the percentage of shares repurchased in the six-month period ending two months after quarter t . We calculate *Repurchase* as -1 multiplied by the percentage change in shares outstanding as reported by CRSP. Thus, the measure is increasing in the number of shares repurchases. We winsorize *Repurchase* to be no less than -3.85% , i.e., if the actual repurchase value is less than -3.85% of shares outstanding, then we replace it with -3.85% . This number represents the 1st percentile in our sample of the distribution of a two-quarter change in treasury stock from Compustat (data #98) deflated by the market value of total shares outstanding at the start of the period.

Ins. Trading is the insider purchase ratio measured as the ratio of the number of shares purchased by insiders in the six-month period ending two months after quarter t , scaled by total number of shares traded (purchases plus sales) by insiders during the same period. Insider trading is limited to open market transactions of the top five executives coded as "CB", "CEO", "CFO", "CO", and "P" by Thomson Financial First Call Insiders Data.

Accruals are computed as the difference between earnings from continuing operations (data #8) in quarter t and cash flows from operations (data #108) in quarter t using data from the statement of cash flow divided by average total assets (data #44) at the end of the quarters $t-1$ and t .

SUE is computed as the difference between the net income (data #69) in the quarter t and the net income (data #69) in the quarter $t-4$ deflated by the market value of equity (data #14 * data #61) in the quarter $t-4$.